Sabancı University
CS310 – Mobile Application Development
2026-2027 Fall Project Step 1 Proposal

SUnite Find your group at Sabancı

Group 32

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GitHub: `https://github.com/ssinix/sunite`

1 App Overview

What the app does: SUnite is a mobile app, exclusive to Sabancı University students, for finding and forming groups of any kind. It is organised as a set of category boards on which students can post and browse listings of two kinds: *existing groups looking for new members*, and *students looking for a group to join*. Boards cover course groups, Erasmus, startups/projects and campus events.

Problem it addresses: Finding the right people at SU is currently scattered across WhatsApp groups, SUCourse forums and word of mouth. A student who needs a project partner for a course, wants to meet others going to the same Erasmus school, is looking for co-founders with a specific skill, or simply does not want to go to a club event alone has no single place to look. SUnite gathers all of these requests in one place and makes them searchable by course, school, skill and event.

Intended users: Sabancı University undergraduate and graduate students, verified through their `@sabanciuniv.edu` e-mail address. Clubs and student initiatives benefit indirectly, since students can organise groups around their events.

2 Core Features

1. **SU-only authentication:** sign up and log in with a university e-mail address, with e-mail verification.
2. **User profiles:** name, e-mail, year of study, major(s), minor(s), skills/specialties and current groups; users can view each other's profiles.
3. **Four category boards:** *Course Groups*, *Erasmus*, *Startups & Projects* and *Events*.
4. **Two post types:** "group looking for members" and "person looking for a group".
5. **Create, edit and delete posts/groups:** title, description, capacity (current/maximum), "looking for" (specific qualities or skills needed) and a category-specific field (course code, Erasmus school and term, or event name and date).
6. **Browse, search and filter:** filter a board by course code, school, tag or groups that still have open spots.
7. **Join requests:** a student requests to join a group; the group owner accepts or declines. Capacity updates automatically and the post closes when the group is full.
8. **My Groups page:** groups the user owns or belongs to, their members and pending join requests.

Optional Features (if time permits)

- **Random matchmaking:** "match me" with other students who are also looking for a group for the same course or event.

- **Friends:** add friends and see which groups they are in.
- **In-app group chat** for members of a group.
- **Push notifications** for new join requests and acceptances.
- **Course list autocomplete** from the SU course catalogue.

3 Data Requirements

The app requires a cloud database so that posts and groups are shared between all users in real time. We plan to use **Firebase Authentication** for accounts and **Cloud Firestore** (NoSQL) for application data. No external API is required for the core features. Small local preferences (e.g. last used filters) will be kept on the device with `shared_preferences`.

Collection	Main fields
`users`	uid, name, e-mail, year, majors, minors, skills, group IDs
`groups`	title, description, category, capacity, owner ID, member IDs, "looking for", category field (course code / school & term / event & date)
`posts`	type (group seeking members / person seeking group), category, author ID, linked group ID, text, created at, status (open/closed)
`joinRequests`	group ID, user ID, status (pending/accepted/declined), created at

Optional: `friendships` (user pairs), `messages` (group chat).

4 Potential Challenges

- **Restricting access to SU students:** enforcing the e-mail domain and verification, and writing Firestore security rules so users can only edit their own data.
- **Data modelling and queries:** Firestore has limits on multi-field filtering, so board filters will need composite indexes and some data duplication.
- **Consistency:** two students may be accepted into the last spot at the same time; membership and capacity updates must use transactions.
- **State management:** keeping boards, group pages and profiles in sync with real-time updates across many screens.
- **Privacy and moderation:** letting users control which profile details are visible and handling inappropriate posts.

5 Team Responsibilities

Step 1 coordinator: Mert Ali İlter

Member	Assigned tasks
Azra At	Login Page and Database
Ece Ulutas	Groups Page and Design
Elif Çamlıkaya	Main Page and Design
Mert Ali İlter	Backend